

Logan Brunet

Game Tools & Engine Engineer · Orlando, FL (Hybrid / Remote / Onsite)

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github.com/lobotomy-x/portfolio

SUMMARY

Senior C++ engineer specializing in game engine tooling, real-time rendering, and immersive experiences. 23 years of C++ experience; deep hands-on work across Unreal Engine, Dear ImGui, C#/WPF, Lua, HLSL/GLSL, and WinAPI. Recent independent work includes IDE-class feature development on a production ImGui editor (22 features shipped), a ground-up WPF injection frontend for the UEVR VR modding platform, and agentic AI-assisted development workflows using Claude Code. Comfortable owning full stacks from engine internals through deployment and developer tooling.

TECHNICAL SKILLS

Languages	C++ (23 yrs), C# (.NET Framework 4.6.2-4.8 / .NET 6+), Python 3, Lua, JavaScript / TypeScript, HLSL, GLSL, Kotlin
Engines	Unreal Engine 4/5, Unity (URP/HDRP), Dear ImGui, ARKit/ARCore
Graphics	D3D11/12, Vulkan, DXGI swapchain hooks, ReShade plugin API, shader authoring
Platform	WPF/XAML, WinAPI P/Invoke, WMI, Sol2, CMake, SDL3, ImGuiFileDialog
AI & Tooling	Claude Code (agentic workflows), LLM API integration, prompt engineering, AI-assisted code pipelines, Copilot, Gemini
3D & Creative	Blender (10k+ hrs, 2 yrs paid), VR-ready modeling, animation rigging, real-time asset pipelines
Tools	Git, Visual Studio, Rider, VS Code, Jira, Confluence, Tortoise SVN, Adobe CC

PROFESSIONAL EXPERIENCE

Game Tools & Modding Engineer · micro1 (Contract) May 2026 · Remote
• Contracted via micro1 to deliver engine tooling and modding work on a proprietary C++ codebase. Details under NDA.

Unreal Game / Simulation Developer · Adacel Jan 2026 – Apr 2026 · Orlando, FL (Hybrid)
• Modernized legacy simulation codebase to modern C++ and Epic internal standards — memory-safe pointer logic, UObject reflection, UPROPERTY/UFUNCTION systems for Blueprint extension.
• Designed and built suite of ImGui-based diagnostic plugins providing real-time telemetry, performance profiling, and runtime debugging for high-fidelity simulation environments.
• Advised technical lead on Unreal Engine architecture, internals, and core gameplay framework decisions.
• Conducted vulnerability audits of proprietary engine assets and serialization protocols; designed IP protection roadmap.
• Built and evaluated AI-based workflows for project setup, IP protection, debugging, and code refactoring.

ImGuiColorTextEdit — IDE-class editor (C++ / Dear ImGui)

- Shipped 22 features on a production multi-platform Dear ImGui code editor — fold engine, IDE navigation, runtime language loading, and bug fixes across macOS, Linux, and Windows.
- Fold engine: Python-aware indent folding, per-type collapse previews, anti-aliased fold arrows, fold-all/unfold-all via VS Code-style Ctrl+K chord shortcuts.
- IDE navigation: header/source toggle (Alt+O), #include go-to-file, tab reopen history, click-drag text move/copy.
- Runtime language loader — new languages added by dropping a config file, no recompile. 6 starter languages shipped. F5 script runner for Python, Lua, JS, Bash, PowerShell.

UEVR Frontend (C# / WPF desktop application)

- Built a system-tray application from scratch that automatically injects VR support into any Unreal Engine game — detects launches via WMI process-start trace, manages per-game profiles, no user interaction required.
- Implemented raw Shell_NotifyIcon tray from scratch (not WPF wrappers) — message-only HWND, WndProc, icon state switching, Shell_NotifyIconGetRect context menu positioning.
- GitHub Releases API auto-updater with local cache, version picker for rollback, and periodic background checks.
- Unelevated Steam/Epic launcher management to prevent anticheat elevation issues; Windows startup shortcut with self-updating target path; single-instance mutex enforcement.

UEVR Lua API (C++ / Sol2 / Dear ImGui)

- Upgraded Dear ImGui across 5 years of upstream versions with full docking support; nearly doubled Lua binding surface (~1900 to ~3500 lines).
- Authored Matrix4x4 and Transform Lua bindings from scratch; implemented Unreal Engine core math in pure Lua (vectors, quaternions, transforms, projection matrices).
- Wrote API documentation and Lua training guide used by modding community; supported and trained 12+ active modders.

ReShade Open Source Contributor

- Implemented text editor features and HLSL shaders for graphics debugging and post-process effects.

3D Artist · Brown Center for Computation and Visualization

Providence, RI

- Created VR-ready 3D models of campus architecture for real-time delivery.

EDUCATION

Brown University

B.S. Cognitive Neuroscience — Computational Cognitive Neuroscience, Statistical Inference, Game Development (Unity), Public Speaking

AP: Computer Science 5 · BC Calculus 5 · Chemistry 5 · National Hispanic Scholar · AP Scholar with Distinction

Bootcamps: Full Stack Developer, Per Scholas (Java, Spring Boot, SQL, JS, REST APIs) · Data Analytics, General Assembly (Excel, Tableau, SQL)

REFERENCES

Travis Clement, Technical Lead, Adacel — tclement@adacel.com · Sonia Mina, Engineering Manager, Adacel — smina@adacel.com